

Nicole Mitchell | nicole.elyse.mitchell@gmail.com

B.S., M.S. Computer Science | Research Scientist, Athlete, Artist, Writer

RESEARCH EXPERIENCE

Google Research, SF — *Federated Learning, Privacy, Unlearning*

SEP 2024 - PRESENT | Research Scientist

Longitudinal Risks

Role: Core contributor

Wrote a position piece advocating for longitudinal studies of human-AI interactions to better understand and mitigate risks that develop over time.

Key accomplishments:

- Position paper submitted to EMNLP 2026 ([preprint](#)).

Privacy Auditing for DP Synthetic Data Generation

Role: Lead investigator, primary contributor

Introduced a new canary design, which leverages high-temperature sampling to yield highly sensitive synthetic probes for detecting privacy leakage, as well as an improved approach for auditing synthetically generated data.

Key accomplishments:

- Carried out implementation and experimentation.
- Paper submitted to EMNLP 2026 ([preprint](#)).

Models that Forget Innovation Pod

Role: Research program lead

Led a research effort leveraging modularity and unlearning to control data influence and address privacy and policy challenges. 1 of 4 proposals selected from 56 submissions to the Athena Innovation Pods program.

Key accomplishments:

- Built team of collaborators, drew connections across orgs, charted out research roadmap, secured buy-in from leadership.
- Proposed data compartmentalization as a strategy for AI Governance, presenting a taxonomy mapping governance needs to technical requirements and relevant strategies. GenLaw@ICML 2024 ([preprint](#)).
- Documented internal use cases for unlearning.
- Hosted 3 student researchers, investigating approximate unlearning for robustness ([preprint](#)) and certifiable unlearning for privacy ([preprint](#)).
- Presented research outcomes to leadership.

MAY 2022 - SEP 2024 | Research Engineer

Fine-tuning LLMs with User-level Differential Privacy

Role: Core contributor

Investigated how to best scale user-level DP training to LLMs under fixed compute budgets.

Key accomplishments:

- Helped define and execute on experiments ([preprint](#)).

Function-Space Aggregation of Models in Federated Learning

Role: Mentor, project lead

Led a student researcher project to explore new algorithms for merging models in federated learning, drawing upon ideas in meta-learning.

Key accomplishments:

- Scoped and drafted a proposal; received funding.
- Implemented federated model training and evaluation infrastructure to enable running large-scale experiments on various algorithms.
- Paper accepted to TMLR 2024 ([preprint](#)).

San Francisco, CA

[nicolemitchell.github.io](https://github.com/nicolemitchell)

[Google Scholar](#) | [LinkedIn](#) | [Substack](#)

ABOUT ME

Thorough and astute researcher with **excellent communication skills**.

Natural leader with proven ability to strategize, draw connections, build collaborations, coordinate large-scale efforts and influence stakeholders.

Research interests span areas of trustworthy and efficient ML with recent work focused on **data privacy and provenance**.

Strong technical background in computer science and software engineering.

Capable of spanning the **research to production** pipeline, and cares about doing so.

Passionate about exploring the people side of AI and **societal implications of technology**.

LEADERSHIP & COMMUNITY CONTRIBUTIONS

Peer-review publications - ICML, TMLR, NeurIPS.

Mentor Student Researchers - provide research and engineering guidance through daily syncs and co-working sessions.

Contribute to open-source - e.g., [TensorFlow Compression](#), [Dataset Grouper](#).

Co-led the Brain Women community and now host the Women in Research @ SFO lunch series - conduct monthly socials, organize talks and plan events.

Helped facilitate [Research Retrospectives](#) - host interviews with established researchers to shed light on their careers.

SELECT PUBLICATIONS

Advancing the State-of-the-Art in Empirical Privacy Auditing N.
Mitchell, G. Andrew, A. Ganesh, H.B. McMahan, P. Kairouz, *under review at EMNLP (2026)* [[link](#)]

Building a Scalable Dataset Pipeline for Group-Structured Learning

Role: Core contributor

Developed infrastructure to support scaling up federated learning simulation to datasets and models orders of magnitudes larger than previously used.

Key accomplishments:

- Released an open source library, [Dataset Grouper](#), for partitioning existing flat datasets into group-structured datasets.
- Paper accepted to NeurIPS Benchmarks & Datasets 2023 ([preprint](#)).

AI Explorable: How Federated Learning Protects Privacy

Role: Project lead

Created an [AI Explorable](#) on [federated learning](#). Through interactive visualizations the piece allows readers to wrestle with data privacy concerns, learning how FL aims to mitigate them while introducing trade-offs.

Key accomplishments:

- Conceived of the project and forged a cross-company collaboration.
- Worked with key stakeholders to incorporate feedback that balanced the narrative with Google's privacy story (spot bonus).
- Crafted a concise, informative, and visually engaging piece that highlighted research areas and key responsibility issues (spot bonus).

FEB 2021 - MAY 2022 | AI Resident

Designing & Implementing a Compression Method for Federated Learning

Role: Lead investigator, primary contributor

Leveraged entropy coding to design a method for compressing model updates in federated learning that optimizes the communication-accuracy trade-off. Deployed this custom compression method in Google's FL production system.

Key accomplishments:

- Implemented [compression-based aggregation methods](#) for federated learning in TensorFlow Federated (TFF). This involved: custom TF ops, TFF logic, and robust integration tests for system compatibility.
- Presented at Google Research Conference.
- Published a [TinyPaper](#) at ICLR 2023; delivered an oral presentation.
- Released [custom compression op](#) as open-source software in TensorFlow Compression.
- Reduced communication 320x with <0.1% drop to model accuracy on GBoard next-word prediction (spot bonus).

EDUCATION

Rice University, Houston — B.S., M.S. Computer Science

AUG 2015 - MAY 2020 | Advised by [Lydia Kavradi](#)

Master's Thesis: Machine Learning-Based Prediction of Sites of Metabolism in Drugs: Exploring Feature Extraction Methods on Molecular Graphs ([link](#))

Activities: NCAA Division I Track & Field, Student Athlete, Baker Institute for Public Policy, Developing Civic Scientist Leaders Program participant ([link](#))

ADDITIONAL PURSUITS

Pacific Crest Trail, Mexico to Canada

APR 2025 - AUG 2025 | Thru-hiker, writer, artist

Took a sabbatical to spend 4 months hiking the Pacific Crest Trail.

Walked a very long way, drew the landscapes I passed through, and wrote about my experience ([Substack](#)).

Long-term Measurements:

Towards a Longitudinal

Understanding of Human-AI

Interactions N. Mitchell, D. Agarwal, M. Bohacek, R. Denton, R. Patel, under review at EMNLP (2026) ([link](#))

Leveraging Function Space

Aggregation for Federated

Learning at Scale

N. Dhawan, N. Mitchell, Z.B. Charles, Z. Garrett, G.K. Dziugaite, TMLR (2024) ([link](#))

Towards Federated Foundation

Models: Scalable Dataset Pipelines for Group-Structured Learning

Z.B. Charles, N. Mitchell, K. Pillutla, M. Reneer, Z. Garrett, NeurIPS (2023) ([link](#))

How Federated Learning Protects

Privacy N. Mitchell, A. Pearce, PAIR AI Explorables (2022) ([link](#))

Optimizing the

Communication-Accuracy

Trade-off in Federated Learning

with Rate-Distortion Theory

N. Mitchell, J. Ballé, Z.B. Charles, J. Konečný, ICLR (2023) ([link](#))

PEER FEEDBACK

"I really like how Nicole is not afraid to stop and ask questions whenever things are unclear or contradictory. It is crucial to be precise in science and research, and the human communication or thought process isn't always, which is why questioning each other and ourselves is a big part of our work."

"Nicole has a high attention to detail, which is fantastic to see. She works independently to understand new concepts and once she has, she delivers technically excellent code."

"Nicole demonstrates independence in exploring what and/or how else we could visualize experimental results to be more insightful."

AWARDS & RECOGNITIONS

Adobe Research Women in Technology Scholarship (2019)

Rice Computer Science Graduate Research Fellowship (2019)

President's Honor Roll (2018)

CRA-W Grace Hopper Celebration Research Scholar (2018)